Qiao Sun

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RESEARCH & PROFESSIONAL EXPERIENCE

Undergraduate Researcher, He Vision Group Sep 2024 - Present

Professor: Kaiming He

- Work as a UROP student in He Vision Group on generative models, including diffusion and flow matching, normalizing flows, and fast image generation.
- Collaborate with Hanhong Zhao, Zhicheng Jiang, Xianbang Wang, and Yiyang Lu on a unified direction for vision-language understanding and text-to-image generation.
- Conduct large-scale experiments with JAX on TPUs and PyTorch on GPUs; build training, evaluation, and ablation pipelines for image generation models.
- Serve as one of the group leads and TPU/GCP managers/admins, helping coordinate compute resources, access, and shared training infrastructure.

Quant Strategy Analyst Intern, Ubiquant Investment (Jiukun Quant) Jun 20 - Aug 31, 2025

Quantitative research and strategy analysis

- Researched quantitative strategy ideas and backtesting workflows with Python-based data analysis.
- Evaluated factor robustness, risk and turnover characteristics, and strategy diagnostics through statistical tests and visualization.
- Built reproducible research utilities for experiment tracking, factor validation, and portfolio-performance analysis.

SELECTED PAPERS

Is Noise Conditioning Necessary for Denoising Generative Models? 2025

ICML 2025 poster; first author

- Revisited the necessity of noise conditioning in diffusion and flow-matching models by reimplementing eight denoising generative models across multiple datasets.
- Proposed uEDM, a noise-unconditional diffusion model with competitive performance, supported by theory that matches empirical behavior.

Bidirectional Normalizing Flow: From Data to Noise and Back 2025

CVPR Spotlight; project lead and first author

- Revisited normalizing flows with a learned reverse map, eliminating explicit inverse-flow computation and slow autoregressive inference.
- Guided reverse learning through hidden-state alignment, enabling single-evaluation NF-based generation with state-of-the-art results among NF methods.

One-step Latent-free Image Generation with Pixel Mean Flows 2026

ICML accepted; first author

- Built a strong baseline for one-step, pixel-space, latent-free generation using MeanFlow with x-prediction.
- Reported 2.22 FID on ImageNet 256 and 2.48 FID on ImageNet 512; open-source implementation at Lyy-iiis/pMF.

COURSE PROJECTS

Fast Humanoid Loco-Manipulation via Flow Matching 2025

MIT 6.4210 course project; robotics manipulation

- Project video: https://www.youtube.com/watch?v=MIpTpM4C71k&t=2s
- Reimplemented a BeyondMimic-style humanoid loco-manipulation pipeline with simulated data preparation, diffusion training, and post-hoc control guidance.
- Replaced DDPM sampling with flow matching and demonstrated lower-latency control with 5 sampling steps.

EDUCATION

Massachusetts Institute of Technology Sep 2024 - Present

Undergraduate in Artificial Intelligence and Mathematics; expected graduation: 2028; GPA: 5.00/5.00

Tsinghua University, Institute for Interdisciplinary Information Sciences Sep 2023 - Jul 2024

Pre-college student; GPA: 4.00/4.00

CERTIFICATES & HONORS

Top 17, 2025 Putnam Mathematical Competition 2025

2nd place, 2024 Putnam Mathematical Competition 2024

Gold Medal and 11th place, International Mathematical Olympiad 2023

Gold Medal and 1st place with perfect score, Chinese Mathematical Olympiad 2022

Excellent Award, Alibaba Global Mathematics Competition; top 70 among 50,000+ participants 2022-2024

SKILLS

Programming
Python, C, PyTorch, JAX, GPU/TPU training and evaluation

Languages
Mandarin (native), English (fluent)